

HW 2: Three-level systems



Question 1) Recall that a two-state system could be represented by a Bloch sphere ($\mathcal{H} = \mathbb{C}^2 \rightarrow \mathbb{CP}^1 \simeq S^2$), that is, a two-dimensional complex Hilbert space maps to a two-dimensional real space once we factor out the unphysical global phase. The two real parameters are usually represented by a polar and azimuthal angles, θ and ϕ , on the Bloch sphere. This describes pure states of spin-1/2 particles ($s = 1/2$, $m_s = \pm 1/2$) or qubits. For generic, mixed states (we will cover those later), the Bloch vector need not lie only on the spherical surface, but also in its interior.

We will now consider a three-state system with basis $\{|+\rangle, |0\rangle, |-\rangle\}$. You can imagine this as a spin-1 particle ($s = 1$, $m_s = -1, 0, +1$), or a “qutrit”. This system now requires a four real number parameterization, $\mathcal{H} = \mathbb{C}^3 \rightarrow \mathbb{CP}^2$. Unfortunately the latter cannot be represented by S^4 as you might be tempted to conclude. In fact, it turns out that to represent a generic (mixed) state in a three-state system, you’d need an 8-dimensional real space (as opposed to a 3-dimensional real space for the two-state system). This homework should convince you that while in 2-state systems we can reconstruct any state from its Bloch vector (three expectation values), in 3-state systems the Bloch vector is not enough, even for pure states. We will need 8 expectation values instead.

Any normalized physical state in a three-state system can be written as

$$|\psi\rangle = c_+ |+\rangle + c_0 |0\rangle + c_- |-\rangle, \quad (1)$$

with

$$|c_+|^2 + |c_0|^2 + |c_-|^2 = 1, \quad (2)$$

and

$$c_+ = \cos \frac{\theta}{2}, \quad c_0 = \sin \frac{\theta}{2} \sin \frac{\theta_0}{2} e^{i\phi_0}, \quad c_- = \sin \frac{\theta}{2} \cos \frac{\theta_0}{2} e^{i\phi}, \quad (3)$$

where $\theta_{(0)} \in [0, \pi]$ and $\phi_{(0)} \in [0, 2\pi)$.

- Show that fixing the polar angle $\theta_0 = 0$ leads to a subspace that recovers the two-state system (Bloch sphere) with $\{|+\rangle, |-\rangle\}$ as the basis kets. What are the polar and azimuthal angles that describe this subspace?
- What choice of angles gives a Bloch sphere with $\{|0\rangle, |-\rangle\}$ as the basis kets? How about $\{|0\rangle, |+\rangle\}$ as the basis kets? What are the polar and azimuthal angles that describe these subspaces? We have found three two-state subspaces within the three-state system, but there are infinitely many other possible two-state subspace combinations.
- Interpret this system as a spin-1 particle (here is where we start disconnecting from qutrits). We can define the x , y , and z components of the spin-1 operator as

$$\begin{aligned} \hat{S}_x &= \frac{\hbar}{\sqrt{2}} (|+\rangle \langle 0| + |0\rangle \langle +| + |0\rangle \langle -| + |-\rangle \langle 0|), \\ \hat{S}_y &= \frac{\hbar}{\sqrt{2}} (-i|+\rangle \langle 0| + i|0\rangle \langle +| - i|0\rangle \langle -| + i|-\rangle \langle 0|), \\ \hat{S}_z &= \hbar(|+\rangle \langle +| - |-\rangle \langle -|). \end{aligned} \quad (4)$$

Find the matrix form of these operators in the basis

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |-\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad (5)$$

Express your answer as a 3×3 matrix as well as in terms of the Gell-Mann matrices, λ_i . These are the 3×3 analogs of the Pauli matrices, generators of the $\mathfrak{su}(3)$ algebra, the algebra of the $SU(3)$ group. Just like any operator could have been written in terms of the Pauli matrices for a two-state system, any operators in a three-state system can be written in terms of the Gell-Mann matrices.

- d) Check that the spin operators in Eq. (4) obey the same algebra as the spin- $\frac{1}{2}$ operators, $[\hat{S}_i, \hat{S}_j] = i\hbar\epsilon_{ijk}\hat{S}_k$, and that $\hat{\mathbf{S}}^2 = s(s+1)\hbar^2\mathbb{1} = 2\hbar^2\mathbb{1}$. While generic three-state system operators are built on the $\mathfrak{su}(3)$ algebra, the spin operators themselves still obey the familiar $\mathfrak{su}(2)$ algebra.
- e) Calculate the Bloch vector

$$\mathbf{S} = \begin{pmatrix} \langle \hat{S}_x \rangle \\ \langle \hat{S}_y \rangle \\ \langle \hat{S}_z \rangle \end{pmatrix} \quad (6)$$

for $|+\rangle$, $|0\rangle$, $|-\rangle$, and the state

$$|s\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle). \quad (7)$$

These are all pure states. What can you say about the length of $\mathbf{S}$ for the $|0\rangle$ and $|s\rangle$ states? How do these two examples contrast with the Bloch vector for the two-state system?

- f) Calculate $\langle \hat{S}_{x,y,z}^2 \rangle$ and $\Delta S_{x,y,z} = \sqrt{\langle \hat{S}_{x,y,z}^2 \rangle - \langle \hat{S}_{x,y,z} \rangle^2}$ for $|+\rangle$, $|0\rangle$, $|-\rangle$, and $|s\rangle$. How does your result for $|0\rangle$ and $|s\rangle$ compare with the previous question? Can one infer the quantum state of the system solely from the Bloch vector for a three-state system? Convince yourself that this is the case for **two**-state systems.



Bonus) In the 2-state system, we needed 2 real numbers to specify a pure state. Any observable can be expressed in terms of the Pauli matrices, and knowing the Bloch vector allows us to determine the expectation value of any observable.

In three-state systems, we need only 4 real parameters to specify states. However, because observables can be expressed in terms of the Gell-Mann matrices, we would actually need to know 8 real parameters to fully determine the expectation values of all observables. The Bloch vector only gives us 3 out of 8. Here you will calculate the remaining 5.

The Bloch vector in three-state system is called the polarization vector. The rest lives in what is called the alignment tensor,

$$T_{ij} = \langle \hat{S}_i \hat{S}_j + \hat{S}_j \hat{S}_i \rangle - \frac{4}{3}\hbar^2\delta_{ij}, \quad i, j = x, y, z. \quad (8)$$

Note this is a symmetric, traceless rank-2 tensor.

- a) Prove this is a traceless tensor, that is,

$$\text{Tr}(T) = T_{xx} + T_{yy} + T_{zz} = 0. \quad (9)$$

This shows that we go from generic $3 \times 3 = 9$ degrees of freedom to 6 after imposing symmetry, and then 5 after imposing tracelessness. With the polarization vector, we have 3 more, recovering the total of 8.

- b) Calculate the components of the alignment tensor T_{ij} for the states $|0\rangle$ and $|s\rangle$.